

Technology Stack

Date	2 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3 (Glassmorphism styling), Bootstrap 5	Enables a highly responsive, modern, visually-stunning user interface for all devices.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.12, Flask Framework	Lightweight backend routing, rapid development, and native compatibility with Python ML libraries.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	JSON Database (crop_stats.json), CSV Raw Data	Directly saves memory footprint, allows zero-setup loading on local servers, and easy parsing in Flask.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Host Server (Port 5001)	Provides a cost-effective, immediate development workspace for testing system functionalities locally.
5	Version Control & CI/CD	Git, GitHub	Maintains version control history, code tracking, and remote backup storage.

	(e.g., GitHub, GitLab, Jenkins)		
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Chart.js, FontAwesome	Chart.js enables rendering high-fidelity interactive analysis graphs; FontAwesome provides modern vector icons.